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VAtools: a python package for customizable annotation of VCF files

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Software

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Summary

VAtools (VCF Annotation Tools) is a package designed to ease manipulation of genomic data stored in the common VCF format, so that annotations can be added or extracted in a user-friendly manner.

Statement of need

The VCF format is the de facto standard for reporting and sharing genomic variant data, but it exists within an ecosystem of bioinformatics tools that output a wide array of often ill-defined formats. In order to store data in a consistent manner and simplify downstream analyses, computational biologists and genomics researchers often need to integrate these results into a VCF file. Doing so with existing toolchains often requires complex multi-step machinations. We built VAtools to simplify some of the most common operations.

State of the field

Several python libraries exist that allow programmatic access to a VCF, such as pysam, VCFPy, and PyVCF (Heger et al., 2026; Holtgrewe & Beule, 2016; Vermaat et al., 2023). Using these generally involves writing custom wrappers and requires detailed knowledge of both the structure of a VCF and of the input data. The VCF format is complex (with a 50-page specifications document as of version 4.5), it can be difficult to parse and manipulate, and this represents a substantial hurdle for many users. Other tools, like bcftools, support general command line manipulation, but rely on the same deep knowledge and have a complicated syntax (Danecek et al., 2021). VAtools was designed to provide a simple interface for annotating VCFs with some of the most common types of genomic data.

Software design

VAtools was built with ease of use and modularity as the primary goals. VAtools was motivated by our own difficulty in preparing inputs for variant analyses, and provides tools that simplify a variety of common annotation tasks. Some are designed specifically around common genomic programs: `vcf-readcount-annotator` adds detailed metrics from bam-readcount, while `vcf-expression-annotator` can link variants to specific gene or transcript expression information from Kallisto or Stringtie, among others (Bray et al., 2016; Khanna et al., 2022; Pertea et al., 2015). In addition, the package provides more general tools that can take arbitrary data with genomic coordinates and incorporate them into a VCF using `vcf-info-annotator` (e.g. dbSNP allele frequencies or ClinVar pathogenicity classifications). It also includes utilities for extracting variant effect annotation data from VCFs into tabular format via `vep-annotation-reporter`, fixing common issues with malformed VCFs with `vcf-genotype-annotator` and for sanity checking of results with `ref-transcript-mismatch-reporter`.

Creating this set of tools installable with a single call to `pip` fits well with the paradigms of genomic analysis, which favors rapid installation and the ability to prototype and iterate via the command line. These prototypes often mature into more complex genomic pipelines, for which modularity and containerization are critical. Distributing via pip and Bioconda enables easy installation and version management, as well as containerization that is critical to modern bioinformatics workflow engines like Toil, Cromwell, or Nextflow (Di Tommaso et al., 2017; Vivian et al., 2017; Voss et al., 2017).

VAtools was built on top of the `VCFPy` library, selected over two alternatives in part because of its intuitive syntax that eases development and code review. `PyVCF` (or its new `PyVCF3` fork) was not used because it lacks convenient accessors for modifying VCF headers and per-sample genotype fields. Another alternative, `pysam`, has faster I/O, but requires strict checking of `##contig` lines in the header. As many genomic pipelines do not produce such fields, we ultimately determined that using pysam would limit the reach of this tool. Though not built on the fastest library, VAtools is nonetheless performant, annotating 1 million records with vcf-info-annotator in 47.8 seconds on a Macbook Pro M2 with 16Gb of RAM and using Python 3.14.0. Further optimizations are possible, but unnecessary for the typical use case of this tool, which operates on the scale of individual exomes (thousands of variants) to whole genomes (~3 million variants).

Research impact statement

VAtools was originally designed to work in concert with pVACtools (Hoang et al., 2026; Hundal et al., 2020), an immunotherapy tool that incorporates many different sources of data (VEP annotations, gene expression from multiple sources, allele specific expression, readcounts). Custom pipelines using VAtools provided the underlying data for neoantigen and immunotherapy studies in pediatric ependymoma, lung cancer, and high-grade serous ovarian cancer (Garsed et al., 2022; Miller et al., 2018; Nicholas et al., 2026). More recently, VAtools components have been incorporated into a pipeline called ImmunoNX (Singhal et al., 2025) that provides an end-to-end workflow for cancer vaccine design. It has also been used to create neoantigen vaccines in clinical trials for triple-negative breast cancer, follicular lymphoma, and glioblastoma (Garfinkle et al., 2026; Ramirez et al., 2024; Zhang et al., 2024), as well as at least 10 other clinical trials currently in progress.

Outside of cancer immunotherapy, VAtools has found use in a variety of other contexts, including importing sequencing metrics to enable filtering of genomic variants in studies of leukemia and malignant peripheral nerve sheath tumors (Abel et al., 2023; Dehner et al., 2021) or in mouse models of skin cancer (Shea et al., 2023). It was also used in evolutionary studies of the amoeba Dictyostelium discoideum (Walker et al., 2024).

We expect that VAtools will continue to be generally useful to the genomics community, enabling applications ranging from large-scale automated pipelines to small ad-hoc workflows. VAtools is released under an MIT license to allow for broad reuse. The code is available at https://github.com/griffithlab/vatools, with documentation at https://vatools.org. Versioned releases are made available on PyPI, GitHub, Docker Hub, Bioconda, and Zenodo (doi:10.5281/zenodo.21079855).

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AI usage disclosure

No AI was used in the first ~8 years of VAtools development, but starting in 2026, we have used Claude Sonnet 4.6 for occasional assistance in adding new features or refactoring. All code (whether generated by humans or AI) has undergone rigorous code review by the authors of the package and each change must pass a comprehensive test suite integrated into the repository.

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